Math 526, S12 **Name:** Answer

Quiz 9 **Section:** **SSN:** xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find ther best fitting line $y = C + Dt$ for $b = 1, 2, 4$ at times $t = -1, 0, 2$.

(a) Write down the resulting equation $\mathbf{Ax} = \mathbf{b}$.

$$Ax = b$$

where

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} C \\ D \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

(b) Solve the normal euqation for the least squares fitting, $\mathbf{A}^T\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^T\mathbf{b}$, to find C and D.

$$A^TA\hat{x} = A^Tb$$

$$A^TA = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 5 \end{bmatrix}$$

$$A^Tb = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix}$$

Solve

$$\begin{bmatrix} 3 & 1 \\ 1 & 5 \end{bmatrix}\begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{cases} C = 2 \\ D = 1 \end{cases}$$

Then the least squares fitting line is

$$y = 2 + t$$

Problem 2. (5 points) Find an orthonormal basis $\mathbf{q}_1$, $\mathbf{q}_2$, $\mathbf{q}_3$ for the column space of $\mathbf{A} = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix}$ using the *Gram-Schmidt* process.

Solution: $\tilde{q}_1 = a_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\|\tilde{q}_1\|^2 = 1^2 + 0^2 + 0^2 = 1$

$\tilde{q}_2 = a_2 - \left(\frac{\tilde{q}_1^T a_2}{\tilde{q}_1^T \tilde{q}_1}\right)\tilde{q}_1$ $\qquad \tilde{q}_1^T a_2 = 2\cdot 1 + 0\cdot 0 + 3\cdot 0 = 2$

$= \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} - \left(\frac{2}{1}\right)\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$= \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$ $\qquad \|\tilde{q}_2\|^2 = 0^2 + 0^3 + 3^2 = 9$

$\tilde{q}_3 = a_3 - \left(\frac{\tilde{q}_1^T a_3}{\tilde{q}_1^T \tilde{q}_1}\right)\tilde{q}_1 - \left(\frac{\tilde{q}_2^T a_3}{\tilde{q}_2^T \tilde{q}_2}\right)\tilde{q}_2$

$= \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} - \left(\frac{4}{1}\right)\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \left(\frac{18}{9}\right)\begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$ $\qquad \tilde{q}_2^T a_3 = 0\cdot 4 + 0\cdot 5 + 3\cdot 6 = 18$

$\tilde{q}_1^T a_3 = 1\cdot 4 + 0\cdot 5 + 0\cdot 6 = 4$

$= \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}$

$\Rightarrow q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} / 1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$q_2 = \frac{\tilde{q}_2}{\|\tilde{q}_2\|} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} / 3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$q_3 = \frac{\tilde{q}_3}{\|\tilde{q}_3\|} = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} / 5 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$